

**Assignment: Reactions and Stoichiometry**
Name: Solomon

Score: \_\_\_\_\_ / 40 marks

Instructions: Write answers in the right-hand column.

Section A—Multiple choice (10 marks)

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 1 | <p>For which of the following reactions would the following observation apply:<br/>two colourless solutions are mixed to form a white precipitate.</p> <p>A a solution of magnesium nitrate is mixed with a solution of sodium sulfate</p> <p>B solutions of potassium carbonate and nickel chloride are mixed</p> <p><u>C</u> dilute hydrochloric acid is added to a solution of silver nitrate <u>AgCl</u></p> <p>D a solution of sodium hydroxide is added to dilute nitric acid</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | C | 2 |
| 2 | <p>The reaction of sodium sulfite with hydrochloric acid may be represented by the equation:</p> $\text{Na}_2\text{SO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g}).$ <p>On the basis of the information in the equation, which one of the following is true?</p> <p><u>A</u> When 1 mol of <math>\text{Na}_2\text{SO}_3</math> is treated with acid, 1 mole of <math>\text{SO}_2</math> will always be produced. ✓</p> <p>B <math>\text{Na}_2\text{SO}_3</math> and <math>\text{H}^+</math> must always be present in the mole ratio of 1:2 for a reaction to take place. ✗</p> <p>C The ratio by mass of <math>\text{Na}_2\text{SO}_3</math> to <math>\text{H}^+</math> in a reaction mixture containing the correct stoichiometric proportions is 1:2. ✗</p> <p>D 0.4 mole of <math>\text{H}_2\text{O}</math> will be produced when 0.8 mole of <math>\text{H}^+</math> reacts completely with excess <math>\text{Na}_2\text{SO}_3</math>. ✗</p> | A | 2 |
| 3 | <p>When solutions containing calcium ions and phosphate ions are mixed, the following precipitation reaction occurs:</p> $3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s})$ <p>How many mole of precipitate would form if solutions containing 0.825 mole of <math>\text{Ca}^{2+}</math> and 0.520 mole of <math>\text{PO}_4^{3-}</math> were mixed?</p> <p><u>A</u> 0.260 mol <u>LR</u></p> <p>B 0.275 mol</p> <p>C 0.520 mol</p> <p>D 0.825 mol</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | A | 2 |

## Assignment: Reactions and Stoichiometry

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 4 | <p>When 0.200 mol of calcium is mixed with 880 g of water, 4.48 L of hydrogen gas form (at STP). How would the volume of hydrogen produced change if the mass of water were to be decreased to 440 g?</p> <p style="text-align: center;"><math>\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca}^{2+}\text{(aq)} + 2\text{OH}^-\text{(aq)} + \text{H}_2\text{(g)}</math></p> <p>A Only one half the volume of hydrogen would be produced.<br/> <b>B</b> The volume of hydrogen produced would be the same. ✓<br/>         C The volume of hydrogen produced would double.<br/>         D No hydrogen would be produced.</p> | B | 2 |
| 5 | <p>In an experiment to determine the empirical formula of a chloride of a metal X, a student reduced 0.277 g of the chloride of X to the metal X itself, by using aluminium powder as the reducing agent. She obtained 0.207 g of X. <span style="color: purple;">∴ 0.07g of Cl</span><br/>         (relative atomic masses: X = 207; Cl = 35)<br/>         From these results, the formula of the compound is shown to be:</p> <p>A XCl<br/> <b>B</b> XCl<sub>2</sub><br/>         C XCl<sub>3</sub><br/>         D XCl<sub>4</sub></p>                                                                                              | B | 2 |

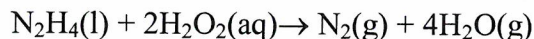
## Section B—Written answers (30 marks)

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 6 | <p>a solutions of iron(III) nitrate and potassium hydroxide are mixed</p> <p>equation:<br/> <math>\text{Fe}(\text{NO}_3)_3 + 3\text{KOH} \rightarrow 3\text{KNO}_3 + \text{Fe}(\text{OH})_3(\text{s})</math></p> <p>observation: Pale Brown Precipitate formed ✓</p> <p>b solutions of calcium hydroxide and sulfuric acid are mixed to give a precipitate of calcium sulphate, and water.</p> <p>equation:<br/> <math>\text{Ca}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}</math></p> <p>observation: No visible reaction ✓</p> | 6 |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

## Assignment: Reactions and Stoichiometry

7

Liquid hydrazine reacts explosively with hydrogen peroxide according to the equation:



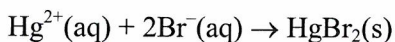
Determine the volume, at STP, of gaseous products that would be formed by the reaction of 3.50 L of 4.30 mol L<sup>-1</sup> hydrogen peroxide with excess hydrazine.

$$\begin{aligned} V &= 3.50 \text{ L} \\ C &= 4.30 \text{ mol L}^{-1} \\ n &= C \times V \\ &= 3.50 \times 4.3 \\ &= 15.05 \text{ mol} \checkmark \\ \therefore n(\text{H}_2\text{O}) &= 2 \times 15.05 \\ &= 30.1 \text{ mol} \\ n(\text{N}_2) &= \frac{15.05}{2} \\ &= 7.525 \text{ mol} \\ \text{Total } n &= 37.625 \checkmark \\ \therefore V &= 22.71 \times 37.625 \\ &= 854.46 \text{ L} \checkmark \\ &= 854 \text{ L (3 Sig Fig)} \end{aligned}$$

3

8

What volume of 0.0712 mol L<sup>-1</sup> ammonium bromide solution will be required to precipitate all of the mercury(II) ions from 33.6 mL of 0.144 mol L<sup>-1</sup> mercury(II) nitrate solution? The ionic equation for the reaction is:



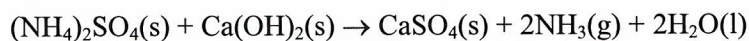
$$\begin{aligned} \text{Hg}^{2+} + 2\text{Br}^{-} &\rightarrow \text{HgBr}_2 \\ \text{Hg}(\text{NO}_3)_2 + 2\text{NH}_4\text{Br} &\rightarrow \text{HgBr}_2 + 2\text{NH}_4\text{NO}_3 \\ n(\text{Br}^{-}) &= 2 \times n(\text{Hg}^{2+}) \checkmark \\ n(\text{Hg}^{2+}) &= 0.144 \times 0.0336 \\ &= 0.004838 \text{ mol} \checkmark \\ \therefore n(\text{Br}^{-}) &= 2 \times 0.004838 \\ &= 0.009676 \checkmark \\ n &= C \times V \\ \therefore V &= \frac{n}{C} \\ &= \frac{0.0096768}{0.0712} \\ &= 0.136 \text{ L} \checkmark \\ &= 136 \text{ mL} \checkmark \\ &= 136 \text{ mL (3 Sig Fig)} \end{aligned}$$

5

## Assignment: Reactions and Stoichiometry

9

Ammonium sulfate and calcium hydroxide react in accordance with the following equation:



A mixture of 109 g of ammonium sulfate and 53.0 g of calcium hydroxide is gently heated, causing the reaction to occur. What mass of ammonia is produced?

$$n((\text{NH}_4)_2\text{SO}_4) = \frac{109}{132.1414} \checkmark$$

$$= 0.82456 \text{ mol} \checkmark$$

$$n(\text{Ca}(\text{OH})_2) = \frac{53.0}{74.086} \checkmark$$

$$= 0.7153 \text{ mol} \checkmark$$

$$\therefore \text{Ca}(\text{OH})_2 \text{ is L.R} \checkmark$$

$$\therefore n(\text{NH}_3) = 2 \times 0.7153$$

$$= 1.431 \text{ mol} \checkmark$$

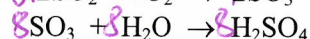
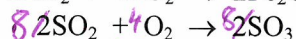
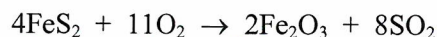
$$m(\text{NH}_3) = 1.431 \times 17.031 \checkmark$$

$$= 24.37 \checkmark$$

$$m = 24.4 \text{ g} \checkmark$$

## Assignment: Reactions and Stoichiometry

- 10 Sulphuric acid can be manufactured from iron pyrites according to the following sequence of reactions:



- a) Calculate the mass of sulphuric acid which could be obtained from 1.00 kg of pure iron pyrite.
- b) If the iron pyrites is only 85% pure, what mass of sulphuric acid could be produced from 1.00 kg of impure ore?

a)  $n(\text{FeS}_2) = \frac{1000}{119.97} \checkmark$   
 $= 8.335 \text{ mol} \checkmark$

$n(\text{H}_2\text{SO}_4) = 2 \times n(\text{FeS}_2) \checkmark$   
 $= 16.67 \checkmark$

$\therefore m(\text{H}_2\text{SO}_4) = 16.67 \times 98.076 \checkmark$   
 $= 1635.095 \checkmark$

$\therefore m = 1.64 \text{ kg} \checkmark$

b)

$0.85 \times 1.64$   
 $= 1.39 \text{ kg} \checkmark$

